

PES UNIVERSITY, Bangalore
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Department of Computer Science & Engineering
UE24CS341A : Software Engineering

LAB - 01

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Problem Statement - 64: Disaster Relief Supply & Volunteer Coordinator

Use-Case Flow Specification

Use Case: Match and Prioritize Shelter Supply Request

Use Case ID: UC-01

Primary Actor: Disaster Manager

Purpose

To match a field shelter's supply request with available warehouse supplies and assign the appropriate delivery priority.

Preconditions

- A supply requisition has been received from a field shelter.
- The requested item and quantity are available in the requisition.
- Warehouse supply information is available to the system.

Postconditions

- A field shelter has already submitted a supply request, and the request is shown as "Pending Review".
- The Disaster Manager is logged in and has permission to manage deliveries.
- The system can connect to the Warehouse System to check the available stock.

Main Success Scenario

1. The Disaster Manager opens the list of pending requests and selects a request.
2. The system checks the type and quantity of supplies requested.
3. The system checks the available stock in the warehouse.
4. The Warehouse System sends the current stock information to the system.
5. The system checks the type of supply and decides its priority.
6. Water and medical kits are given Critical priority, while blankets and other shelter items are given High, Medium, or Low priority as required.

7. The system shows the suggested priority and the amount of stock available to the Disaster Manager.
8. The Disaster Manager confirms the priority.
9. The system adds the request to the logistics manifest and changes its status to "Prioritized".
10. The use case ends successfully.

Alternate Flow – Insufficient Supply

1. The Warehouse System shows that there is not enough stock to fulfil the complete request.
 2. The system marks the request as "Partial Fulfillment" and increases its priority by one level.
 3. The system shows the Disaster Manager how much stock is available and the shortage amount. It also shows an alternate warehouse if available.
 4. The Disaster Manager can either:
 - Approve a partial delivery, or
 - Wait until the required stock is available.
 5. The process then continues with the updated quantity.
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